Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 07-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Subcutaneous Emicizumab in Participants From Birth to 12 Months of Age With Hemophilia A Without Inhibitors **Official Title:** A Phase IIIb, Multicenter, Open-Label, Single-Arm Study to Evaluate the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Subcutaneous Emicizumab in Patients From Birth to 12 Months of Age With Hemophilia A Without Inhibitors **Short Title/Acronym:** HAVEN 7 **Protocol Number:** MO41787 (EudraCT: 2020-001733-12) **NCT Number:** NCT04431726 **Sponsor Name:** Hoffmann-La Roche (Roche) **Investigational Product:** Emicizumab (Hemlibra) **Phase of Development:** Phase 3 (Phase IIIb) **Trial Status:** Active, not recruiting **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of emicizumab prophylaxis in infants based on annualized bleeding rate (ABR) endpoints for treated bleeds using model-based, mean calculated, and median calculated estimates. **Secondary Objectives:** To evaluate the safety, pharmacokinetics (PK), and pharmacodynamics (PD) of emicizumab in infants (≤12 months), track joint health and MRI scores, and monitor the development of adverse events.

2.2 Background & Rationale To address the gap in clinical data regarding the use of early routine prophylaxis in infants with severe Hemophilia A. Subcutaneous emicizumab enables prophylaxis from birth, potentially reducing the high risk of life-threatening bleeding and intracranial hemorrhage (ICH) associated with early infancy. HAVEN 7 is the first trial dedicated to infants designed to confirm that early initiation of emicizumab is efficacious, safe, and can alleviate the intense treatment burden (e.g., intravenous central lines) of standard FVIII replacement.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target N: 55 male participants (infants) **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 04-Feb-2021 (Actual) **Estimated Primary**

Completion: 22-May-2023 (Actual) **Current Status:** Active but not currently recruiting

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age from birth to ≤ 12 months at the time of informed consent. 2. Body weight ≥ 3 kg at the time of informed consent. Premature babies (gestational age <38 weeks) may be enrolled provided they reach ≥ 3 kg and corrected gestational age is reported. 3. Mandatory receipt of vitamin K prophylaxis according to local standard practice. 4. Diagnosis of severe congenital hemophilia A (intrinsic FVIII level < 1%). 5. Negative test for FVIII inhibitor (< 0.6 Bethesda units [BU]/mL) assessed during screening. 6. No history of documented FVIII inhibitor, FVIII drug-elimination half-life < 6 hours, or FVIII recovery < 66%. 7. Previously untreated patients (PUPs) or minimally treated patients (MTPs; ≤ 5 days of exposure to hemophilia-related treatments). 8. Documentation of the details of hemophilia-related treatments and bleeding episodes received since birth. 9. For patients from birth to < 3 months of age: no evidence of active intracranial hemorrhage (ICH), confirmed by a negative cranial ultrasound at screening. 10. Adequate hematologic, hepatic, and renal function. 11. Parent/caregiver willingness and ability to comply with protocol requirements.

4.2 Exclusion Criteria 1. Inherited or acquired bleeding disorder other than severe hemophilia A. 2. Use of systemic immunomodulators (e.g., interferon) at enrollment or planned use. 3. Prior use of emicizumab prophylaxis, an investigational drug to treat hemophilic bleeds within 5 half-lives, or non-hemophilia investigational drug within the last 30 days (or 5 half-lives). 4. Current active severe bleed (e.g., ICH) or planned surgery (excluding minor procedures like circumcision or CVAD placement). 5. History of clinically significant hypersensitivity associated with monoclonal antibody therapies. 6. High risk for thrombotic microangiopathy (TMA) or previous/current treatment for thromboembolic disease. 7. Any hereditary or acquired maternal condition that may predispose the patient to thrombotic events, or other diseases increasing bleeding/thrombosis risk. 8. Known infection with HIV, hepatitis B virus, or hepatitis C virus; or serious infection requiring antibiotics/antivirals within 14 days prior to screening. 9. Unwillingness of the parent/caregiver to allow receipt of blood or blood products. 10. Concurrent disease or lab abnormality precluding safe participation.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Loading Dose:** 3 mg/kg once weekly (QW) for 4 weeks. * **Maintenance Dose (Year 1):** 3 mg/kg once every 2 weeks (Q2W) for 52 weeks. * **Long-Term**

Follow-Up (Years 2-8): Participants may continue 3 mg/kg Q2W, or switch to 1.5 mg/kg QW, or 6 mg/kg once every 4 weeks (Q4W) over a 7-year long-term follow-up period. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** 52 weeks of primary treatment phase, followed by a 7-year long-term follow-up period (Total 8 years).

5.2 Concomitant & Prohibited Therapy Permitted: Mandatory receipt of vitamin K prophylaxis according to local standard practice. Treatment of breakthrough bleeds with FVIII products as per on-demand clinical guidelines. **Prohibited:** Systemic immunomodulators, investigational agents, and concurrent prophylactic bypass agents unless specified by the protocol.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent by Caregiver, Medical History, Cranial Ultrasound (<3 mo), Inhibitor Labs, Weight check
Loading Phase	Weeks 1-4	SC Injections (QW), PK/PD baseline assessments, Caregiver education
Maintenance	Weeks 5 to 52	SC Injections (Q2W), Safety Labs, Efficacy Assessment (Bleed logs)
Long-Term Follow-up	Years 2 to 8	Maintenance dosing (QW, Q2W, or Q4W), AE tracking, Joint Health/MRI evaluation

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Model-Based Annualized Bleeding Rate for Treated Bleeds:** Estimated ABR over the efficacy period using a negative binomial regression model (incorporating the 72-hour rule for bleeds at the same anatomical location). 2. **Mean Calculated Annualized Bleeding Rate for Treated Bleeds:** Calculated as (number of bleeds / number of days during the efficacy period) x 365.25. 3. **Median Calculated Annualized Bleeding Rate for Treated Bleeds:** Standardized formula based on treated bleed incidence (excluding surgery/procedure bleeds).

7.2 Secondary & Exploratory Endpoints 1. **Hemophilia Joint Health Score (HJHS):** Total score at specified timepoints during the long-term follow-up period. 2. **Magnetic Resonance Imaging (MRI) Score:** Assessment of specific joints at specified timepoints during the long-term follow-up period. 3. **Safety & Toxicity:** Incidence and severity of Adverse Events (AEs), with severity determined according to the World Health Organization (WHO) Toxicity Grading Scale.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via electronic caregiver diaries and regular clinician evaluations, graded per the WHO Toxicity Grading Scale. Monitored strictly for local injection-site reactions and systemic events. **Serious Adverse Events (SAEs):** Specific monitoring for TMA, thromboembolic events, and ICH; requiring immediate reporting. **Data Monitoring Committee (DMC):** Independent safety monitoring mechanisms to oversee trial data, particularly given the vulnerable infant population.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy analysis is conducted on the Intent-to-Treat (ITT) population who received at least one dose. Safety Set for all AE evaluations. **Sample Size Justification:** $N=55$ provides sufficient clinical and pharmacokinetic data to characterize the safety, efficacy, and steady-state exposure in the infant population. **Handling of Missing Data:** ABR calculations incorporate exposure days. The 72-hour rule applies (two bleeds of the same type/location within 72 hours counted as one). Model-based imputation is applied for missing covariate data where appropriate.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study is conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular on-site and remote monitoring to ensure accurate recording of bleed events, dose compliance, and centralized lab sample processing. **Audit Trail:** eCRF locking, tracking of protocol deviations (e.g., missed doses), and centralized documentation of FVIII inhibitor testing.

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11. Study Identifiers & Governance

Primary ID: {{MO41787}} **Registry Number:** {{NCT04431726}} **Sponsor/ Lead Organization:** {{Roche (Hoffmann-La Roche)}} **Study Title:** {{HAVEN 7}} **Official Regulatory Title:** {{A Phase IIIb, Multicenter, Open-Label, Single-Arm Study to Evaluate the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Subcutaneous Emicizumab in Patients From Birth to 12 Months of Age With Hemophilia A Without Inhibitors}}

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12. Clinical Framework

Primary Condition: {{Severe Hemophilia A}} **Diagnosis Threshold:** {{Intrinsic FVIII level < 1%}} **Medical Methodology:** {{Subcutaneous Emicizumab Prophylaxis}} **Optimization Target:** {{Zero treated bleeds and prevention of intracranial hemorrhage (ICH)}}

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13. Study Procedures & Methodology

Management Pathway: {{Emicizumab maintenance dosing algorithm}} **Education Protocol (HCP):** {{Subcutaneous injection training and pediatric bleed management}} **Education Protocol (Patient):** {{Caregiver training for SC injection, bleed recognition, and diary logging}} **Quality Audit Mechanism:** {{Centralized assessment of FVIII inhibitors and PK/PD parameters}}

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14. Observation & Follow-Up Schedule

Total Duration: {{416 weeks (8 years)}} **Onsite Visit Cadence:** {{Screening, Weeks 1-4, then periodic visits per Q2W/Q4W schedules}} **Remote Monitoring Frequency:** {{Continuous via caregiver electronic bleed diaries}} **Checklist Review Interval:** {{Routine clinic visits for PK/PD and immunogenicity assessments}}

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | | Clinical Operations Lead | [Name] | ___ |** ***[DD-MMM-YYYY]* | | *Lead Statistician* | *[Name]* |** _____ | [DD-MMM-YYYY] |